MadLoop with TIRust

One-loop reduction · registered campaign report

CutTools · Ninja · COLLIER · NinjaRS · TIRust

Starting MadLoop revision `9228fc58e8f8` · full helicity/color sums

Selected campaign: `split-dp-terminal-1`. Current numerical identity: `accb7e50eec50093`.

Registered exporter anchor: `b33801545f13`. Observed paired-timing export revisions: not yet measured. Documentary revisions can differ while the registered exporter bytes remain identical.

Complete delivery qualification: incomplete. 10 of 11 physical cases are closed: 60 lanes × 100 saved points pass 3e-6. EW n4 remains pending in this frozen snapshot. The eleven-process numerical matrix, controls, existing-backend regressions, precision cohorts, implementation checks and unresolved stress cases retain separate statuses.

Precise performance acceptance: deferred to Hua-Sheng. Final timings will run on his local machine. Shared-host timings are provisional, without CPU reservation, and cannot establish the ≤1.05 upper confidence bound. This snapshot contains no new paired timing samples.

Runtime colour: green = faster, white = parity, red = slower. All displayed ratios are candidate / reference; smaller is faster.

Table 1: Main process matrix. Timings include the actual rescue work. Physics and runtime acceptance are separate.

Process · correction	Physics	Baseline µs/point	TIRust chain µs/point	Ratio · 95% interval	Existing-backend regressions
$u\bar{u} \to Z$ QCD · 2→1 · 100 points	pass	—	—	— deferred	deferred
$gg \to t\bar{t}$ QCD · 2→2 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to Zgg$ QCD · 2→3 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to ZZgg$ QCD · 2→4 · 100 points	pass	—	—	— deferred	deferred
$u\bar{d} \to W^+$ EW · 2→1 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-$ EW · 2→2 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to t\bar{t}\gamma$ EW · 2→3 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ EW · 2→4 · 100 points	pending	—	—	— deferred	deferred

This study measures the new integration against the untouched starting branch, which already includes NinjaRS and earlier Fortran optimizations. It does not attribute those earlier improvements to TIRust.

1 Retained predecessor failures

The following qualified failures belong to earlier candidate implementations. They remain authoritative for those numerical sources and fixed policies. Results for the selected campaign appear in the main matrix; a new campaign does not replace or erase a previous candidate's failure.

Table 2: All retained qualified ancestor failures, including discovered and unselected attempts. Source identities and every read evidence file are bound in the report build manifest.

Process · comparison	Previous candidate source	Retained ratio · 95% interval	Retained failure evidence
$u\bar{u} \to Z$ `tirust-first`	`legacy` revision `c91cb85f3aa9` source SHA `74116e4bb2ba7abd`	1.12105 1.10349–1.13889 **fail**	`runs/qcd_n1_uux_z/tirust-first-attempt-3`

2 Five-backend numerical agreement

The same saved momenta, model parameters, coupling orders and scale are evaluated by all five tools. The regression criterion is `3e-6`, with `1e-8` as the ordinary-point target. Born, finite, simple-pole and double-pole coefficients are retained. Near-zero poles use the existing MadLoop comparison policy. Final accepted DP/QP outcomes and failed evaluations are retained; the histograms do not enumerate internal rescue attempts.

This table reparses the final complete steady warmup traversal: 100 saved points, baseline CutTools and five candidate reducers. It is untimed numerical evidence. It supplies neither paired runtime acceptance nor native-DP cohort timings. Failure flags and internal retry counters are **unobserved**.

Process · saved points	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to t\bar{t}$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$u\bar{u} \to Zgg$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 87 / QP 13 1e-8: miss
$u\bar{u} \to ZZgg$ main · 100 expected	**pass** DP 88 / QP 12 1e-8: pass	**pass** DP 97 / QP 3 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 96 / QP 4 1e-8: miss	**pass** DP 14 / QP 86 1e-8: miss
$u\bar{d} \to W^+$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to \mu^+\mu^-$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to t\bar{t}\gamma$ main · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 82 / QP 18 1e-8: pass
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ main · 100 expected	**pending**	**pending**	**pending**	**pending**	**pending**
$u\bar{u} \to g\gamma$ control · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to H$ control · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to HH$ control · 100 expected	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass

Every lane is compared with the baseline CutTools rows on exactly matching point and split-order indices. The separate 1e-8 target is not substituted for the 3e-6 regression criterion. Failed, incomplete and corrupt observations retain their diagnostics and never become successful rows.

Agreement is assessed using independent numerical backends and higher precision when a discrepancy occurs. Known difficult scalar-box regions are retained as stress cases. A backend's own success flag does not establish cross-backend agreement.

3 Completed physical pilot checks

These eight separately retained checks each evaluate 100 saved physical points with all five reducers. They use the common strict stability policy and pass the `3e-6` regression criterion. Their timings are ineligible for performance acceptance, and they do not fill the pending timing-campaign rows above. An independent audit reparsed 4,800 evaluations and checked the retained inputs, executable provenance and raw statuses.

These historical pilots retain their original candidate sources in the pilot index. They do not establish numerical acceptance for the selected campaign `split-dp-terminal-1`.

Table 3: Historical physical pilots, independent of the runtime gate. Residuals exclude poles accepted by the documented small-pole exception. All four existing reducers finally accepted 100 DP results in each row.

Process · 100 points	Five-backend regression	Ordinary target	Largest symmetric residual	TIRust accepted DP / QP
$u\bar{u} \to Z$	pass	pass	2.69×10^{-13}	100 / 0
$gg \to t\bar{t}$	pass	pass	4.58×10^{-12}	100 / 0
$u\bar{u} \to Zgg$	pass	miss	2.81×10^{-6}	87 / 13
$u\bar{d} \to W^+$	pass	pass	1.95×10^{-12}	100 / 0
$e^+e^- \to \mu^+\mu^-$	pass	pass	2.65×10^{-12}	100 / 0
$u\bar{u} \to g\gamma$	pass	pass	2.61×10^{-11}	100 / 0
$gg \to H$	pass	pass	1.55×10^{-10}	100 / 0
$gg \to HH$	pass	pass	4.61×10^{-9}	100 / 0

QCD 2→3 adjudication. The original comparison against CutTools DP misses the `1e-8` target at 28 points. CutTools, Ninja and TIRust QP agree within `9.89e-11` on four diagnostic points. Across all 28 target misses, the saved TIRust and COLLIER results meet the target against CutTools QP; CutTools DP misses all 28, while Ninja and NinjaRS miss only point 39. Every comparison still passes the regression criterion. These fixed-QP diagnostics are explicitly unchecked and do not establish automatic rescue stability.

The index `experiments/numerical/completed-pilots-index.json` binds 491 retained files and preserves the original comparisons. The detailed QP adjudication is in `experiments/qcd-n3-adjudication/`. Final accepted precision does not enumerate every internal reduction attempt. The remaining three large-process pilots are outside this frozen eight-case index.

4 Absolute timing by reduction backend

Values below are warmed full-amplitude times in µs per point. Initialization, compilation and model generation are excluded and recorded separately. All tools use matching pole, helicity/color and stability settings. Cache behaviour and actual rescue work remain part of the measurement. Recorded backend/precision histograms identify the final accepted reducer for every timed amplitude; they do not count every internal rescue attempt. Control-process speeds are descriptive; the ≤1.05 runtime gate applies to the eight main processes.

Table 4: Per-backend timing observations. Missing measurements are not represented as zero.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

5 Initialization and accepted precision

Each cell gives the first-amplitude initialization plus complete-stream warmup time in seconds, followed by the fraction of timed amplitudes finally accepted in QP. Process startup and parameter-card loading are outside the cold interval. A DP-only label refers to accepted amplitudes; it does not assert that no unsuccessful internal attempt occurred. The warmed timings on the previous page include all rescue work.

Table 5: Cold intervals and final accepted precision. Exact backend/precision counts and point-level results accompany the raw observations.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

6 Full versus poles-only tensors

10 of 12 predeclared settings completed seven alternating full/poles-only timing pairs; two DP settings were refused in both modes. All returned requested pole coefficients agree. These warmed tensor diagnostics use the historical release, with numerical caches disabled. They do not qualify full-amplitude performance.

Table 6: Median wall µs per call and geometric mean paired ratios. Green favors poles-only; red favors full evaluation. Intervals use seven paired log ratios and Student-t with six degrees of freedom.

Tensor · N / rank	Route	Prec.	Full µs	Poles µs	Poles / full · 95% interval
Massive tadpole N=1 · rank 6	Public Fortran	DP	9.01	9.42	**×1.041** 1.032–1.049
Massive tadpole N=1 · rank 6	Public Fortran	QP	1828	1838	**×1.005** 0.999–1.011
Massive bubble N=2 · rank 4	Generated adapter	DP	11.8	11.3	**×0.951** 0.939–0.964
Massive bubble N=2 · rank 4	Generated adapter	QP	1738	1121	**×0.646** 0.644–0.649
Infrared triangle N=3 · rank 3	Generated adapter	DP	8.95	8.78	**×0.977** 0.954–1.000
Infrared triangle N=3 · rank 3	Generated adapter	QP	1038	858	**×0.823** 0.815–0.832
Massive box N=4 · rank 4	Generated adapter	DP	32.2	18.0	**×0.558** 0.551–0.564
Massive box N=4 · rank 4	Generated adapter	QP	15384	2220	**×0.144** 0.141–0.146
Massive pentagon N=5 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive pentagon N=5 · rank 2	Generated adapter	QP	39078	2177	**×0.056** 0.055–0.057
Massive hexagon N=6 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive hexagon N=6 · rank 2	Generated adapter	QP	84852	5144	**×0.061** 0.060–0.061

Scope and observed limits. The rank-six tadpole calls the public Fortran bridge; the other tensors use the existing generated QCD 2⟶4 adapter, whose rank capacity is five. No generated rank limit was changed. The DP tadpole is about 4% slower in poles-only mode; QP is near parity. The massive N=5 and N=6 fixtures have zero poles, so their large QP gains do not describe infrared-singular pentagons or hexagons. DP refusals carry no timing claim.

Numerical caches are off; structural plans and scratch remain warm. Each measured batch lasted 0.825–1.925 s on pinned CPU 9. Driver tracing and coefficient output are outside the timer; input restoration, adapter mapping and bridge serialization remain inside. Shared-host compilation was active; host-load snapshots and CPU times are retained. These timings are explicitly ineligible for the full-amplitude acceptance gate.

Requested poles match within the declared DP/QP bounds; the largest scaled residual is `3.47e-18`. Omitted finite slots are NaN in the generated adapter and zero placeholders in the public bridge. The historical DSO is pinned by SHA256 `49af12f0b882…cd71ca12` and subsequent builds are outside this measurement. Raw batches, exact identities, and the independently audited 79-file inventory are retained under `experiments/poles-timing/`.

7 Methods and qualification

7.1 Physical samples and timing protocol

The main matrix contains QCD and electroweak corrections at every multiplicity from 2⟶1 through 2⟶4, using `loop_sm` and `loop_qcd_qed_sm_Gmu`. Each main case declares 100 deterministic points. On-shell 2⟶1 points vary their boost at the physical invariant; higher multiplicities use 1 TeV. Mixed QCD/QED and loop-induced cases supplement the main matrix. Threshold, soft/collinear, small-Gram and complex-mass cases are separate stress controls.

Performance acceptance uses at least seven alternating paired repetitions. The minimum batch target is one second; calibration aims for two seconds to accommodate shared-host speed variation. Actual durations are retained, and every batch must reach 95% of the minimum. The starting revision and candidate share their compiler, library installations, model/cards, point order and physical settings. Timing begins after the campaign's builds finish and uses one allowed CPU. The upper 95% paired ratio bound must be ≤1.05 separately for every main process. Averages cannot hide a failing row. A FULL–SKIP quantity, when reported in supporting data, is an **estimated reduction stage** rather than a pure tensor-kernel stopwatch.

All timing attempts are retained. Only an objectively short series with passing numerical checks can be retried within the same candidate campaign. A qualified runtime failure remains authoritative; a later favorable repetition cannot erase it. Source optimizations require a new candidate identity and campaign.

The loop-induced controls retain their generated default chain `7|6|1` their prescribed TIRust-first chain is `9|7|6|1`. These are configured chains, not measurements of which reducer ultimately supplied a result.

The recorded cold interval includes first-amplitude initialization and warming the complete saved point stream; it excludes process startup and parameter-card loading. Numerical objects and linked archives are bound to the retained compiler recipes and hashes. The benchmark runs on a shared host; host-load observations accompany timing pairs, and this study does not claim exclusive use of the machine.

The common benchmark card sets `MLStabThres=1e-6` for both revisions and all reducers, with independent stability checks enabled. This policy was frozen before acceptance measurements after a native stress audit exposed correlated errors below the original threshold. Production retains its `1e-3` default. The tighter estimator does not replace the independent numerical truth gate or guarantee detection of every error.

7.2 Interface, poles and caches

MadLoop selects TIRust explicitly as `MLReductionLib=9`. A Fortran adapter transfers supplied invariants, complex squared masses and the squared renormalization scale to a separate Rust bridge. Tensor moments are permuted into MadLoop order and contracted in Fortran. The quadruple-precision path preserves binary128 inputs and uses MPFR working precision; double-double arithmetic is not labelled quadruple precision.

`LoopPoleMode` selects bare-loop UV/IR residues while preserving the ordinary finite coefficient. The distinct poles-only diagnostic path omits finite scalar roots and finite tensor arithmetic; it preserves double-pole contributions to simple poles and scaleless UV/IR separation. Partial results cannot be used as complete physical amplitudes. Cache and precision stores are separated by evaluation mode, and stability rechecks use independent values.

> Detailed interface, cache, QP, poles-only, and relocation qualification records have not yet been incorporated in this data release. Their absence is not a passing result.

7.3 Earlier measured Fortran change

The selected optimization computes ordered momentum prefixes once, retaining the original zero seed and addition order. For six propagators it reduces prefix additions from 60 to 20. At each of `-O0` and `-O3`, 481,536 comparisons passed bitwise output and IEEE-exception checks across DP, QP and four rounding modes. Isolated six-propagator kernel median paired ratios were 0.443/0.347 (DP/QP, `-O0`) and 0.703/0.356 (`-O3`). These are shared-host microbenchmarks, not physical-amplitude speedups. Full-wrapper measurements were smaller and less consistent. Raw observations and source are retained in `experiments/fortran-prefix/`.

7.4 Retained numerical limitations

The bridge's 66 DP/QP stress evaluations include 14 explicit DP refusals and one served DP tensor coefficient above the diagnostic `3e-6` criterion. All 33 QP evaluations pass. For the failing massive rank-five box, native MadLoop reversal checks at the default `MLStabThres=1e-3` can accept correlated DP errors. At `1e-6`, the actual production rescue block with real tensor callbacks switches to QP and resolves the discrepancy below `1.2e-15`. The fixture, failing components,

native check values and QP adjudication remain under `qualification/tirust-bridge/stability-audit/`. This isolated control-flow test omits physical helicity/filter initialization and does not replace matrix-element comparisons.

Pole-only tests prove that finite evaluators and finite tensor arithmetic are skipped. This does not imply every scalar topology is faster: illustrative warm scalar measurements include slower simple massless triangles and boxes, alongside faster massive and soft examples. A completely collapsed massless scalar box returns an explicit error. No unsupported request is counted as accelerated execution.

7.5 Reproduction and provenance

The accompanying `campaigns/split-dp-terminal-1/cases.json` fixes the matrix and thresholds; `campaigns/split-dp-terminal-1/summary.json` is derived from retained observations. The selected manifest is `campaigns/split-dp-terminal-1/manifest.json`. Its ancestor summary snapshots and qualified failures retain the previous candidates' identities separately. The source plan, baseline manifest, dependency revisions, build environment, and raw measurement paths accompany this report. Rebuilding the PDF does not rerun numerical measurements.

Component	Recorded environment
Baseline	`9228fc58e8f8e48dadc70a6fcfb6617845aaeb1f`
Rust	rustc 1.97.1 (8bab26f4f 2026-07-14) (built from a source tarball)
Fortran	GNU Fortran (GCC) 15.2.0
Typst	typst 0.15.1 (9dfd3a08)
TIRust	`854b79dcb7fffcb35e92b4a8a3dd443369b798c4`
oneloopmaster	`bd703d09a78f7849a6b86a07096e6f222fc01193`

Presentation follows the pyAmpliCol report at revision 7266c1600f6e: its page geometry, fonts, navy headings, compact tables and logarithmic timing palette. All numerical measurements and conclusions in this report belong to the present MadLoop integration.

8 Timing eligibility and control-process scope

TODO — Hua-Sheng, local machine. Rebuild the delivered candidate and baseline with identical pinned inputs; replay all 100 saved points; then run at least seven alternating single-core pairs with approximately one-second batches. Report initialization and actual backend/precision use separately. The upper 95% ratio bound ≤1.05 must be checked for every main chain and existing-backend comparison. Shared-host work is best effort without reserving CPUs; it cannot pass this final performance requirement.

User amendment: `PLAN-AMENDMENT-2026-11-10.md`.

Formal main-process timing uses a 2 s calibration target and the frozen 0.95 s minimum observed batch duration. Each batch traverses all 100 saved points. All valid attempts remain visible; a qualified poor result cannot be replaced by a favorable retry. These limits are distinct from the separate traced-cohort diagnostic duration rule.

Main process	Min. baseline s	Min. candidate s	Retained paired attempts
$u\bar{u} \to Z$	—	—	0 · 0 short
$gg \to t\bar{t}$	—	—	0 · 0 short
$u\bar{u} \to Zgg$	—	—	0 · 0 short
$u\bar{u} \to ZZgg$	—	—	0 · 0 short
$u\bar{d} \to W^+$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-$	—	—	0 · 0 short
$e^+e^- \to t\bar{t}\gamma$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	0 · 0 short

Minima include all successfully reparsed retained timing attempts and comparison labels, including ineligible short attempts. Missing or invalid observations do not supply a duration. Selected runs and each attempt's exact minima, labels and eligibility remain in the dossier.

8.1 Descriptive controls

Mixed QCD/QED and loop-induced controls require numerical qualification. Their speed measurements are descriptive and do not carry the main-process ≤1.05 runtime gate. Generic raw-summary failure labels can include slow controls; the role-specific interpretation below preserves those labels without turning control speed into a main acceptance failure.

Control	Paired numerical evidence	Raw summary status	Runtime gate
$u\bar{u} \to g\gamma$	**pending**	pending	not applicable
$gg \to H$	**pending**	pending	not applicable
$gg \to HH$	**pending**	pending	not applicable

The earlier untimed numerical table is a separate dataset. It can be complete before any paired control run exists.

9 Native-DP and rescued cohort diagnostics

Traced full-amplitude subset diagnostics. DP includes internal method selection; compensated DP and naturally rescued QP are separate. Per-call tracing adds overhead. Different frozen subsets cannot be used as backend speed ratios. The diagnostic 90% duration rule is distinct from the main paired 95% rule.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	pending	pending	pending	pending	pending
$gg \to t\bar{t}$	pending	pending	pending	pending	pending
$u\bar{u} \to Zgg$	pending	pending	pending	pending	pending
$u\bar{u} \to ZZgg$	pending	pending	pending	pending	pending
$u\bar{d} \to W^+$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-$	pending	pending	pending	pending	pending
$e^+e^- \to t\bar{t}\gamma$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	pending	pending	pending	pending	pending
$u\bar{u} \to g\gamma$	pending	pending	pending	pending	pending
$gg \to H$	pending	pending	pending	pending	pending
$gg \to HH$	pending	pending	pending	pending	pending

Cells give frozen point counts and diagnostic median µs per call. Empty cohorts have no timing. Short batches, migrations or failure flags cannot yield a valid median. The complete original stream is warmed before membership is frozen; QP timings include failed DP work and natural rescue.

10 Registered implementation and delivery status

Required criterion	Observed status
all eleven five backend numerical	**pending**
eight main runtime gates	**deferred to Hua-Sheng**
existing backend regressions	**deferred to Hua-Sheng**
Installed ABI / Fortran: recorded test groups	**passed**
installer metadata hardening	**open**
native dp and rescued cohorts	**pending**
paired campaign overall	**deferred to Hua-Sheng**
Current statistics round trip	**passed**
stress accuracy	**open**
Subprocess/event cache lifecycle	**passed**

Installed package. DSO `eb3d67fdc99b742b` Fortran archive `0c86349a4fd0dadd` actual installer source `059bb9c97919`. This is distinct from the registered exporter anchor and each observed export HEAD.

The public Fortran parity replay contains 65 calls per installation. The raw DP replay contains 90 calls per library, with 80 successes and 10 expected failures. Cache transport rejection is checked against the retained old-library negative control. These checks qualify the recorded interface and tested transport/cache rejection behavior; their elapsed times are not physical-amplitude benchmark samples.

Integration work remains open. The installer version-type correction is tested only in an isolated module and is not yet applied to production. Adaptive-precision and effective-point changes still need actual backend 8/9 physical qualification. Passed rows describe their recorded test scope; they do not mean all implementation audits are closed.

Subprocess/event cache lifecycle: passed. Four fresh workers each evaluate 19 checked native-DP physical calls and 24 separate DP/QP tensor-key requests across two subprocess prefixes. Coefficients, status, accuracy, inputs and IEEE state agree exactly with caching enabled/disabled and observation enabled/disabled. Each observed worker records 38 automatic physical-boundary clears and two context generations. QP coverage is confined to tensor-key diagnostics; this N≤3, rank≤2 experiment makes no physical-QP-rescue, rank-six, EW, poles-only or timing claim.

`experiments/subprocess-cache-lifecycle/attempt5/qualification-summary.json`

Current statistics: passed. One retained invocation passed 10 tests and four compiled synthetic Fortran tally/writer subtests, using current production Python readers. Legacy aliases, TIRust/NinjaRS precision counters and aggregation are covered. This qualifies the identified statistics sources; it neither recovers the historical unbound 92-pass invocation nor establishes amplitude, native-reducer or performance results.

`experiments/statistics-roundtrip-current-1/manifest.json`

Higher-rank terminal Fortran qualification is bound to the completed source, IEEE/trap and fixed-input coefficient checks in `experiments/dp-higher-rank-production/validation-summary.json`. Its compiled checks and CutTools parity do not substitute for the registered five-backend campaign or establish runtime admission frequency.

11 Retained accuracy limitations and diagnostics

The report binds the completed portable inventories and their recorded scope; it does not rerun scientific adjudication. Historical library results and private prototypes do not qualify the current installed library. Ordinary numerical and paired runtime gates remain separate. The QCD n3 optional target and higher-precision workstream are diagnostic disclosures; neither adds a stress acceptance requirement.

Case	Status	Qualification limit
EW threshold mass transport	**open**	Historical 49af, private mass-transport prototype. QP squaring of existing MP masses reduces the recorded group-5 Gram discrepancy from 1.018% to 1.60e-20, while both arms still refuse 37 threshold tensor parents. Status 120 and accuracy −1 are unchecked; the TIRust sum is incomplete. Matching successful scalar/rank-one subsets does not qualify the failed tensors, full amplitude or performance. `experiments/ew-qp-mass-prototype/observer-native-design/attempt2/execution-1/geometry-audit/README.md`
QCD soft/collinear prototype	**open**	Historical 49af, six private replays. Topology correction plus Born evaluation at repaired coordinates gives NinjaRS status 322 and a 1.33e-15 analytic double-pole error; topology-only status 322 still misses 3e-6. Finite/single-pole truth remains unavailable. TIRust retains 51 nonsingular conditioning refusals and failed status 420; its incomplete sums are not amplitudes accepted for physics. `experiments/qcd-topology-rescue-native/RESULTS.md`
Default stability estimator	**open**	Current eb3d replay now confirms the historical synthetic limitation: all four 1e-3 projections accept inaccurate DP results (maximum finite symmetric error 7.03e-4); all four 1e-6 projections rescue to QP (maximum symmetric error 5.97e-16). This is a synthetic host-helper fixture, not a physical amplitude. The production default remains unchanged; the registered qualification policy is 1e-6. `experiments/default-stability-current-1/README.md`
QCD n3 ordinary target	**open**	Current eb3d campaign: all 100 points pass 3e-6, while the same 28 finite-term 1e-8 misses against CutTools remain. Retained independent QP comparisons support a CutTools DP reference bias and a smaller Ninja/NinjaRS discrepancy at point 39; a reducer cluster alone is not truth. Those historical fixed-QP results are unchecked status 120, not an automatic-rescue certificate. Original comparisons and acceptance gates are unchanged. `campaigns/split-dp-terminal-1/experiments/qcd-n3-target-triage-1/README.md`
Complex-mass extremes	**open**	Nineteen moderate points agree; original extreme automatic-policy points remain unqualified. Private corrections are unchecked diagnostics. `experiments/complex-mass/README.md`
QCD higher precision	**diagnostic**	A separate private all-512-bit replay completed 232 tensor calls with actual runtime mapping and exact ordinary-point parity. It changes arithmetic precision and its admission gate jointly. Forty-six contracted tensor slots retain relative discrepancies, despite close final projections. It does not qualify the adaptive 256-to-512 library or general physical accuracy. `experiments/qcd-topology-rescue-higher-precision-native2/README.md`

The historical c91 qualified runtime failure remains an ancestor result with its original source and interval. New source candidates do not erase it. Ordinary-point agreement and a passing eight-process runtime gate alone cannot declare the full integration goal complete.

12 Experimental adaptive-precision bridge

The experimental bridge is an isolated development build, whose source and evidence are recorded in this public feature branch. It has not replaced the installed TIRust bridge used for the ordinary campaign. The branch is `madloop_tirust` the intended future PR target is `madloop_ninjars`.

12.1 Experimental adaptive-precision bridge

An isolated development build of the TIRust bridge tries 256-bit MPFR first and retries once at 512 bits only for a narrowly identified conditioning refusal. Fortran inputs remain genuine 113-bit QP. Twenty Rust tests pass, including real full/poles tensor kernels and injected failure/cache controls. The separate shared library is built; these tests did not load it through Fortran. Mixed-tier physical amplitudes, dynamic transport, IEEE/trap behavior and performance remain unqualified. It is not installed as the production backend used in this campaign.

`experiments/adaptive-qp-rescue-private-tests/README.md`

12.2 Born/effective-point consistency

Sixteen private exports compile; 51 controlled Fortran checks pass. Twelve ordinary real/CMS legacy runs preserve every compared coefficient exactly. Sixteen fixed-QP runs establish the requested Born/coordinate view, including 24 exact full-arm observations; status 120 is unchecked. No backend 8/9, retained stress amplitude or performance qualification follows from this milestone.

`experiments/born-effective-point-native/README.md`

12.3 Installer metadata hardening remains open

The production installer still accepts boolean values as integer ABI/schema versions; two regression tests remain deliberately failing while the campaign source is frozen. An isolated two-guard correction passes 27 installer/card tests and two extra float-version rejection checks. It has not been applied to production. The passed installed-ABI row covers its identified earlier test groups, not all integration audits.

`experiments/strict-metadata-prototype/README.md`

12.4 QCD n4 target misses

All six 100-point lanes pass 3e-6. TIRust accepts 14 points in DP and 86 in QP. The 1e-8 target is missed at 45 distinct points by at least one candidate reducer; original discrepancies and all pairwise comparisons remain retained. No independent finite/single-pole truth or higher-precision adjudication is supplied by this read-only triage.

`campaigns/split-dp-terminal-1/experiments/qcd-n4-target-triage-1/README.md`

Compilation, mock tests, unchecked fixed-QP calls and a separate all-512-bit experiment each have limited scope. None establishes adaptive-library physical accuracy or final runtime acceptance.